

TeleClinic Platform

Month 3 Learning Review

A full data quality audit, metrics analysis, equity assessment, clinical governance dashboard, learning brief, and north star reflection — covering 1,200 patients, 1,114 consultations, and seven data tables across 12 Rwandan districts.

PLATFORM

TeleClinic · National Telemedicine

PERIOD

February – April 2026 (Month 3)

FUNDERS

Gates Foundation · Ministry of Health · RSSB

RECORDS ANALYSED

1,200 patients · 7 tables · 12 districts

TOOLS USED

Python (pandas) · SQL · Excel · Code Agents

TOTAL POINTS AVAILABLE

100 pts across 5 parts + reflection

CONTENTS

P1	Data Quality Audit — 7 issues, join problem, safety concern
P2	Metrics & Analysis — 5 platform performance metrics
P3	Equity Analysis — CPI, uninsured deficit, Gates brief
P4	Dashboard — Clinical Governance Committee
P5	Learning Review Brief — Gates Foundation
R	Reflection — North Star Metric
N	Process Note

Data Quality Findings

Systematic review of all seven tables. Each issue is characterised by what it is, the most likely explanation, and what it makes unreportable or unreliable.

01 · 307 Consultations with Null Clinical Fields

CRITICAL · DENOMINATOR TRAP

WHAT IT IS

307 of 1,114 consultation rows (27.6%) have simultaneous nulls in `call_type`, `diagnosis_category`, `notes_entered`, and `icd_code_entered`. All 307 share exactly this null pattern.

MOST LIKELY EXPLANATION

These are No-Show (166) and Cancelled (141) appointments. Clinical fields are only populated for completed consultations. This is expected architecture — the system correctly leaves them blank when no clinical encounter occurred.

WHAT IT MAKES UNREPORTABLE

Any analyst using 1,114 as the denominator for ICD compliance, diagnosis mix, or documentation rates will systematically understate performance by ~28%. All clinical metrics must use 807 (completed) as the base.

02 · 213 Patients with Incomplete Registration (17.8%)

HIGH · ACCESS BARRIER

WHAT IT IS

213 patients have `registration_complete = No` and `insurance_validated = No` in perfect lockstep (100% concordance). None of these 213 appear in the Consultations table.

MOST LIKELY EXPLANATION

Insurance verification failed at registration. Failure reasons are nearly equal across four causes: NID not found (28%), insurance expired (25%), invalid format (24%), API timeout (23%).

WHAT IT MAKES UNREPORTABLE

Billing, cost-recovery, and effective coverage rates are unreliable. Whether these patients were denied care or simply haven't yet sought it is unknown. Insurance validation rate calculation requires careful handling.

03 · ICD Codes Missing in 20.9% of Completed Consultations

HIGH · CLINICAL QUALITY

WHAT IT IS

169 of 807 completed consultations (20.9%) have `icd_code_entered = No`. Notes missing in 122 (15.1%). GPs have lower ICD compliance (77.2%) than nurses (83.9%) — a reversed quality gradient.

MOST LIKELY EXPLANATION

Workflow friction: clinicians close the call before completing the record. System may time out logging the consultation as complete before the ICD entry step. Differential UX between GP and Nurse interfaces likely explains the reversal.

WHAT IT MAKES UNREPORTABLE

Disease surveillance incomplete for 169 encounters. Insurance claims for those consultations cannot be reliably processed. Epidemiological trend data for Month 3 is structurally understated.

04 · 62 Uploaded Lab Results Never Reviewed by Clinician

CRITICAL · PATIENT SAFETY

WHAT IT IS

Of 350 successfully uploaded lab results, 62 (17.7%) show `clinician_viewed = No`. Mean time since upload: 41 hours. Affected panels include Blood Glucose, Full Blood Count, and Malaria RDT. Additionally, 23 of 373 ordered tests (6.2%) were never uploaded at all.

MOST LIKELY EXPLANATION

The 23 not-uploaded results reflect lab facility transmission failures (offline or paper-based facilities). The 62 unreviewed results reflect absent result-arrival notifications, or consultation closed before results arrived with no re-engagement trigger.

WHAT IT MAKES UNREPORTABLE

TAT is incalculable for the 23 missing results. Most critically, 62 patients may have had actionable results (positive malaria, elevated glucose) that triggered no clinical response. This is a direct patient safety pathway.

05 · 18 Referrals with Null Authorisation Status (17.6%)

MEDIUM · CARE CONTINUITY

WHAT IT IS

18 of 102 referrals (17.6%) have `authorised = NaN` — neither Yes nor No — with null processing_hours and authorisation_datetime. Diagnoses include Malaria, Diabetes, Mental Health.

MOST LIKELY EXPLANATION

Referrals submitted but not yet processed by the authorising body at time of data extract. They represent a pending queue, not entry errors. However, duration in queue is unknowable without a submission timestamp.

WHAT IT MAKES UNREPORTABLE

True referral authorisation rate (67.6%) is provisional. True denial rate is unknown. Mental Health patients in this pending group may be waiting for escalated care with no outcome recorded.

06 · 491 Registered Patients with No Consultation (40.9%)

MEDIUM · ENGAGEMENT GAP

WHAT IT IS

709 of 1,200 registered patients (59.1%) have at least one consultation. 491 (40.9%) never booked. Of these, 278 completed registration and 213 did not.

MOST LIKELY EXPLANATION

Three sub-populations: (a) recently registered, not yet needed care; (b) patients who registered then disengaged; (c) the 213 blocked by insurance validation

failure. Groups are indistinguishable without booking-attempt logs.

WHAT IT MAKES UNREPORTABLE

True utilisation rate is ambiguous (59.1% of all registered, or 71.8% of fully registered). Dormant vs. never-converted patients cannot be separated without intent/attempt data.

07 · 81 Prescriptions Not Dispensed (15.5%)

LOW-MED · PHARMACY

WHAT IT IS

81 of 521 prescriptions (15.5%) have `dispensed = No` with null `dispensed_datetime` and null `lag_hours` . No dispensing lag can be computed for this cohort.

MOST LIKELY EXPLANATION

Three possibilities: prescriptions pending at pharmacy not yet dispensed at extract time; stockout at designated pharmacy; patient did not collect (non-adherence). Data does not distinguish these scenarios.

WHAT IT MAKES UNREPORTABLE

True dispensing rate is uncertain. For chronic disease patients, undispensed prescriptions represent a continuity-of-care risk. Pharmacy performance metrics are biased by inclusion/exclusion of this group.

Join Problem · Low-Volume District · Patient Safety

The Join Problem — Musanze District Volume Anomaly

FINDING

The Patients and Consultations tables join cleanly on `patient_id` — there are zero orphan records, so this is not a referential integrity failure. The join problem is **logical**: Musanze has 21 registered patients but only 6 consultations (utilisation rate: 28.6%), against a platform average of 63.8%. No other district falls below 57.6%. All 6 Musanze consultations link to valid `patient_ids`. The discrepancy is structural, not a broken key.

HOW I FOUND IT

I left-joined Patients to Consultations on `patient_id`, then computed per-district: registered patients, consultation count, and utilisation rate (completed ÷ registered). Musanze's 28.6% was an extreme outlier — the next lowest district (Gisagara) shows 57.6%. I cross-checked registration dates (spread Feb-Apr 26, not a batch-load issue), channels (mixed USSD/IremboApp), and consultation weeks (only weeks 9-13 represented, suggesting late activation) to confirm the pattern is structural.

HOW I HANDLED IT

Musanze records are **included in all platform-level aggregates** — they are valid data and should not be dropped. However, Musanze is **excluded from any district-level performance rankings or geographic benchmarking** because its anomalously low volume makes per-district rate calculations unreliable and non-comparable. All 5 completed Musanze consultations count toward platform totals.

DISTRICT	PATIENTS	COMPLETED	UTILISATION
Rwamagana	124	97	78.2%
Rulindo	64	46	71.9%
Gasabo	233	157	67.4%
Kicukiro	133	90	67.7%
Huye	136	83	61.0%
Gisagara	33	19	57.6%
Musanze 	21	5	23.8%

Three Explanations for Musanze's Implausibly Low Volume

1. LATE DISTRICT ROLLOUT

Musanze may have been activated mid-quarter. Patients were registered but the clinical workflow (clinician assignment, booking, notifications) was not yet live. Consultations cluster only in weeks 9-13, consistent with a late-quarter go-live. **Additional data needed:** platform go-live date by district; IT deployment log; first booking-attempt timestamp for Musanze patients.

2. CONNECTIVITY / INFRASTRUCTURE GAP

Musanze (Northern Province, near Volcanoes NP) may have poor or intermittent mobile coverage, causing USSD sessions to drop before consultation booking completes. Patients register via SMS but cannot reliably complete a call. **Additional data needed:** call drop-rate by district; network operator coverage maps; channel-specific session abandonment logs.

3. NO CLINICIAN ASSIGNED

If the platform routes by district but no clinician was available to serve Musanze patients, bookings would fail silently or re-route — removing consultations from the Musanze count. **Additional data needed:** clinician-district assignment table; failed/unfilled appointment logs; patient-initiated booking attempts that did not result in a confirmed slot.

Most Concerning Issue — Patient Safety

62 UPLOADED LAB RESULTS NEVER REVIEWED — DIRECT PATIENT HARM PATHWAY

The most urgent patient safety concern is the 62 lab results that were successfully uploaded to the platform but never opened by a clinician (17.7% of uploaded results, mean age 41 hours). For a platform managing malaria, diabetes, and hypertension — all conditions where a single lab value can precipitate an emergency — this is a direct, specific pathway to missed diagnoses and absent treatment escalation. Unlike the ICD coding gap (administrative) or incomplete registration (access), an unreviewed positive malaria RDT or critically elevated blood glucose requires no further investigation to understand its consequences. The platform's architecture currently permits consultation records to be closed without a mandatory result-review step. This is a workflow design failure, not a data quality failure, and it is the single issue I would

escalate immediately to clinical leadership — not schedule for a quality improvement cycle.

Five Key Metrics — Platform Performance at Month 3

Selected to span the full care pathway: access, clinical quality, patient safety, longitudinal engagement, and platform health. Each tells a story the others cannot.

01

CLINICAL QUALITY

ICD Code Completion Rate (Completed Consultations)

79.1%

DEFINITION & CALCULATION

Numerator: Completed consultations where `icd_code_entered = Yes` (n=638).

Denominator: All consultations with `status = Completed` (n=807).

Exclusions: No-Show (166) and Cancelled (141) — no clinical encounter, cannot have ICD.

Result: $638 \div 807 = 79.1\%$ · GPs: 77.2% · Nurses: 83.9%

WHY THIS METRIC? WHAT STORY DOES IT TELL?

ICD compliance is the backbone of disease surveillance, insurance billing, and clinical continuity. No other metric reveals whether a consultation produced structured medical knowledge or an undocumented exchange. The GP vs. Nurse reversal (GPs worse) is the dataset's most actionable clinical quality signal — it points to a specific, fixable workflow intervention rather than a system-wide retraining campaign. At 79.1%, 1 in 5 completed encounters is lost to surveillance and claims processing.

02

PATIENT SAFETY

Lab Result Clinician Review Rate

82.3%

DEFINITION & CALCULATION

Numerator: Records where `result_uploaded = Yes` AND `clinician_viewed = Yes` (n=288).

Denominator: Records where `result_uploaded = Yes` (n=350).

Exclusions: 23 tests where results were never uploaded (lab transmission failure, outside clinician control).

Full pipeline rate (ordered → viewed): $288 \div 373 = 77.2\%$

WHY THIS METRIC? WHAT STORY DOES IT TELL?

Most platforms track TAT. This metric goes further: it asks whether results that arrived were actually used. A result that uploads but is never opened delivers no clinical value. For this disease mix (malaria, diabetes, hypertension), an unreviewed result is not a data gap — it is a patient at risk. The 17.7% non-review rate is the single most alarming finding in the dataset and reveals a workflow design flaw that no reporting metric can fix — only a mandatory acknowledgement step can.

03

PLATFORM ACCESS

Consultation Completion Rate

72.4%

DEFINITION & CALCULATION

Numerator: Consultations with `status = Completed` (n=807).

Denominator: All consultations regardless of status (n=1,114).

No-show rate: $166 \div 1,114 = 14.9\%$ · **Cancellation rate:** $141 \div 1,114 = 12.7\%$

Rural: 74.5% · **Urban:** 69.5% · Gap: +5.0pp (rural outperforms)

WHY THIS METRIC? WHAT STORY DOES IT TELL?

Completion rate is the platform's fundamental access measure. A patient who books but doesn't complete receives no care. The rural-urban equity reversal is striking: rural patients complete consultations at a *higher* rate than urban ones. This challenges the assumption that rural users face more friction. One hypothesis: urban patients have more substitute options (walk-in clinics, pharmacies) while rural patients treat their booked slot as a scarce resource. This nuance is invisible from registration numbers alone.

04

PATIENT ENGAGEMENT

Chronic Disease Follow-Up Rate (DM · HTN · MH)

26.2%

DEFINITION & CALCULATION

Numerator: Follow-up records linked to completed consultations with diagnosis $\in$ {Diabetes, Hypertension, Mental Health} (n=58).

Denominator: Completed consultations with those diagnoses (n=227: DM 77, HTN 94, MH 56).

By condition: Diabetes 29.9% · Hypertension 20.2% · Mental Health 28.6%

WHY THIS METRIC? WHAT STORY DOES IT TELL?

Telehealth for chronic disease only delivers value through longitudinal care. A single consultation for hypertension with no follow-up is not meaningfully better than no care at all. A 26.2% follow-up rate means 3 in 4 chronic patients received a transaction, not care management. For hypertension — Rwanda's leading stroke and cardiovascular risk factor — a 20.2% follow-up rate is a programme design failure. No other metric reveals whether the platform is functioning as a point-of-care tool or a care management system. This one does.

05

PLATFORM HEALTH

Insurance Validation Success Rate

82.3%

DEFINITION & CALCULATION

Numerator: Patients where `insurance_validated = Yes` (n=987).

Denominator: All registered patients (n=1,200).

Failure breakdown: NID not found 27.8% · Insurance expired 25.1% · Invalid format 24.4% · API timeout 22.8%

Retry rate: 150 patients (12.5%) required ≥ 2 attempts

WHY THIS METRIC? WHAT STORY DOES IT TELL?

Insurance validation is the platform's gateway — patients who cannot validate do not receive consultations. The 17.7% failure rate is a direct measure of access denial at the front door. The failure breakdown is the key insight: 22.8% of failures are API timeouts — a *fixable technical problem*. A single engineering change (retry logic, async validation) could resolve ~22% of current failures at zero policy cost. No other metric exposes this specific, immediately actionable intervention opportunity.

What Cannot Be Assessed · Most Concerning Result

Area of Platform Performance That Cannot Be Assessed

PATIENT OUTCOMES — THE PLATFORM CAN SHOW IT IS SEEING PATIENTS. NOT THAT IT IS HELPING THEM.

This dataset captures every operational step — registration, consultation, lab, pharmacy, referral — but contains no health outcome data. There is no measure of whether a patient's hypertension was controlled at follow-up, whether a prescribed antibiotic resolved an infection, whether a referred patient was admitted or turned away, or whether a mental health patient's condition improved. Without outcome data, it is impossible to distinguish a high-performing platform (patients get better) from a high-volume platform (patients are processed). Month 3 data can describe the platform's machinery. It cannot evaluate its purpose.

What is missing: Post-consultation health status assessments (even a simple structured clinician-assessed outcome at follow-up); discharge summaries from receiving referral facilities; repeat chronic disease markers (blood pressure, HbA1c) at follow-up; adverse event or unplanned readmission flags; patient-reported outcome measures.

Diagnosis Mix — Completed Consultations (n=807)

DIAGNOSIS CATEGORY	COMPLETED	% SHARE	FOLLOW-UPS	REFERRALS	REFERRAL RATE
Respiratory Infection	169	20.9%	16	33	19.5%
Malaria	127	15.7%	21	8	6.3%
Hypertension	94	11.6%	19	22	23.4%
Diabetes	77	9.5%	23	14	18.2%
Skin Condition	75	9.3%	9	4	5.3%
Reproductive Health	72	8.9%	9	1	1.4%
Gastrointestinal	68	8.4%	7	2	2.9%
Mental Health	56	6.9%	16	11	19.6%
General Consultation	48	5.9%	6	—	—
Injury / Trauma	21	2.6%	4	7	33.3%

Most Concerning Single Result

62 LAB RESULTS UPLOADED BUT NEVER REVIEWED · 17.7% NON-REVIEW RATE

This is the result I would escalate to clinical leadership before the end of business. An uploaded-but-unreviewed lab result is a clinical non-event: the patient derives no benefit. The platform currently allows consultation records to be completed and clinicians to disengage without any mandatory result-review trigger. For a disease mix that includes malaria (where a positive RDT requires immediate treatment escalation), diabetes (where a blood glucose of 400 mg/dL requires urgent intervention), and hypertension (where a renal function panel may indicate dangerous end-organ damage), a 17.7% silent failure rate is not an acceptable steady state. Unlike the ICD coding gap or registration failures, this is not an administrative shortcoming — it is a structural pathway to patient harm at a platform that has been operating for only three months. If left unaddressed as volume grows to thousands of consultations per month, the absolute number of patients with unreviewed results will scale proportionally.

Equity Assessment — How Equitably Is the Platform Reaching Its Population?

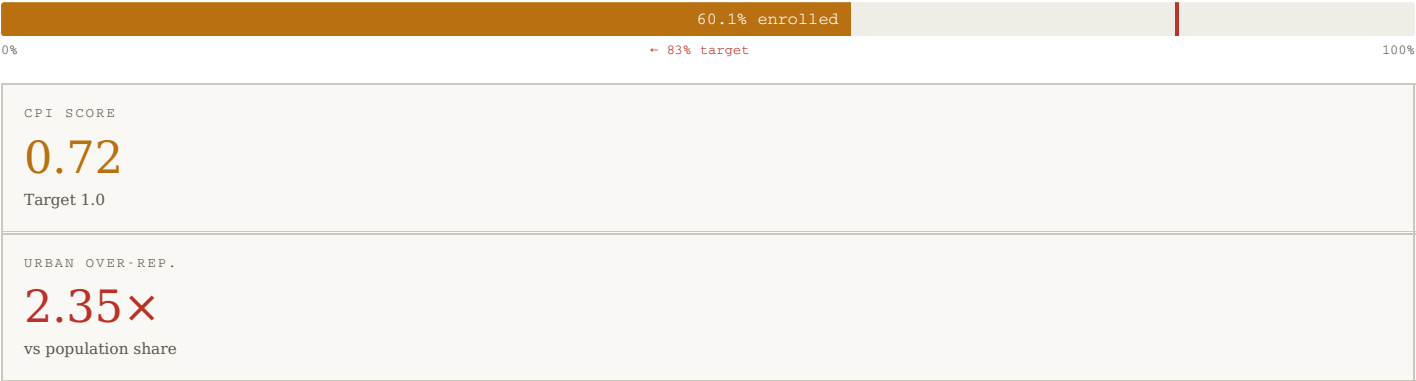
Assessed across four dimensions: geography (rural/urban), channel (USSD vs IremboApp), gender, and insurance status. Two formal equity measures calculated against national benchmarks.

Equity Measure 1 — Rural Coverage Parity Index

Formula: $\text{CPI} = (\text{Rural share of registrations}) \div (\text{Rural share of national population})$
Inputs: 721 rural patients $\div$ 1,200 total = 60.1% · National rural population = 83%
Result: CPI = 0.724 — rural populations are enrolled at 72.4% of their expected rate. A score of 1.0 = perfect proportional representation.

The 22.9 percentage-point shortfall means urban patients are over-represented by a factor of **2.35x** relative to their population share. Despite this registration gap, once enrolled rural patients complete consultations at a *higher* rate than urban patients (74.5% vs 69.5%), indicating that barriers concentrate at enrolment, not at platform use.

Rural share enrolled vs national benchmark (83%)



Equity Measure 2 — Uninsured Consultation Completion Deficit

Formula: Insured completion rate – Uninsured completion rate
Inputs: Insured: 790 $\div$ 1,081 = 73.1% · Uninsured: 17 $\div$ 33 = 51.5%
Result: –21.6 percentage-point gap

Uninsured patients complete fewer than 1 in 2 consultations they book. But the deeper finding is upstream: only 157 patients (13.1% of registrants) are coded as Uninsured, while Rwanda's uninsured rate is substantially higher. 81.5% of patients who registered as Uninsured failed validation (vs. ~8% for insured schemes), meaning **~128 uninsured individuals were likely blocked at the registration door** before ever booking a consultation.

INSURANCE SCHEME	VALIDATION RATE	COMPLETION RATE
RAMA	92.5%	77.0%
MMI	89.9%	75.5%
RSSB-CBHI	90.7%	73.6%
RSSB-Formal	94.2%	70.4%
Uninsured	18.5%	51.5%

Additional Dimensions — Channel & Gender

CHANNEL

Both USSD and IremboApp users are roughly 60% rural (USSD 58.6%, IremboApp 61.1%) — channel choice does not map onto a rural/urban divide. Completion rates are close: IremboApp 73.6%, USSD 70.9%. USSD — the lower-tech channel — does not dramatically underperform, a positive equity signal for low-digital-access populations.

GENDER

Registration is near-balanced: male 50.2%, female 49.8%. Male completion rate is modestly higher (74.1% vs 70.8%, –3.3pp). The significant finding: females are referred at a 4.1pp higher rate (23.0% vs 18.9%) despite near-identical ICD compliance. With only 1 reproductive health referral recorded, this gap has no obvious diagnosis-mix explanation and warrants clinical audit.

“ At Month 3, TeleClinic has enrolled 1,200 patients across 12 Rwandan districts, but the platform's reach is not yet equitable. Rural patients — who represent 83% of Rwanda's national population — account for only 60% of current registrations, a Coverage Parity Index of 0.72 against the national benchmark. Encouragingly, once enrolled, rural patients complete consultations at a higher rate than urban patients (74.5% vs 69.5%), which tells us that barriers are concentrated at the registration and access stage, not in the quality of the platform experience itself. The starkest equity gap lies at the intersection of insurance status and access: the 213 patients (17.8%) whose insurance could not be validated are effectively excluded from care, and uninsured patients complete only 51.5% of booked consultations — a 21.6 percentage-point deficit against their insured peers. This is not a platform failure in isolation: approximately one-quarter of insurance validation failures are attributable to API timeouts in the national verification system, a technical issue that is directly addressable without any policy change. We recommend that the programme immediately implement a provisional access pathway — time-limited consultations processed under manual review — for patients whose insurance validation fails due to system timeouts, to prevent a fixable infrastructure error from becoming a permanent access barrier for Rwanda's most economically vulnerable patients.

— For inclusion in Gates Foundation TeleClinic Quarter 1 Report, June 2026

Most Important Unmeasured Equity Dimension

SOCIOECONOMIC STATUS / POVERTY QUINTILE — THE UBUDEHE DIMENSION

This dataset has no variable capturing household income, poverty status, or socioeconomic stratum. Rural/urban and insurance scheme are imprecise proxies. The data cannot answer the most important equity question: are the poorest Rwandans using this platform at all?

Proposed capture method: Rwanda's national Ubudehe poverty classification (categories 1–4) assigns socioeconomic tiers to households, linkable via NID — which is already queried at the insurance validation step. A secondary Ubudehe API call at registration would retrieve the patient's poverty category with no patient-facing change, adding a single field to the patient record. This would transform equity monitoring: enabling analysis of whether Ubudehe 1–2 households are being reached, whether they face higher validation failure rates, and whether consultation completion is socioeconomically stratified. Without this, the platform's equity narrative rests entirely on geographic proxies.

Clinical Governance Committee Dashboard

Central question: *Is the platform delivering safe, quality care — and for whom?* Designed for a single screen, A4 page. Each element is annotated with design rationale below.

TeleClinic · Clinical Governance Dashboard

▲ 2 ACTIVE SAFETY ALERTS
Review required before next committee meeting

Month 3 · February - April 2026 · 807 completed consultations · 12 districts · Data as of end of Week 13

▲ SAFETY SIGNALS — REQUIRE IMMEDIATE CLINICAL LEADERSHIP REVIEW

● LAB RESULTS NOT REVIEWED

62 of 350 uploaded results (17.7%) never opened by clinician. Mean age: 41h. Includes Blood Glucose, Malaria RDT, Full Blood Count.

● LAB UPLOAD FAILURES

23 of 373 tests ordered (6.2%) — no result uploaded. Lab facility transmission gap. No TAT calculable for this cohort.

● REFERRALS PENDING AUTH

18 referrals (17.6%) neither approved nor denied. Include Diabetes, Mental Health. Duration in queue: unknown.

QUALITY & DOCUMENTATION — CARE PATHWAY COMPLIANCE

CONSULTATION COMPLETION

72.4%

807/1,114 · No-show 14.9%

ICD CODE COMPLIANCE

79.1%

638/807 · GP 77.2% · Nurse 83.9%

LAB RESULT REVIEW RATE

82.3%

288/350 uploaded · 62 not reviewed ▲

RX DISPENSING RATE

84.5%

440/521 · 81 not dispensed

CHRONIC DISEASE FOLLOW-UP RATE — KEY INDICATOR FOR CARE CONTINUITY

DIABETES

29.9%

23/77

HYPERTENSION

20.2%

19/94 — lowest ▲

MENTAL HEALTH

28.6%

16/56

No platform target set — benchmark required for Month 4

REFERRAL PIPELINE — 102 REFERRALS ISSUED

Authorised 67.6% (69)

Denied 17.9% (15)

Pending ▲ 17.6% (18)

Mean auth processing: 24.4h · Max observed: 118h

EQUITY — FOR WHOM IS THIS PLATFORM WORKING?

RURAL COVERAGE PARITY INDEX

0.72

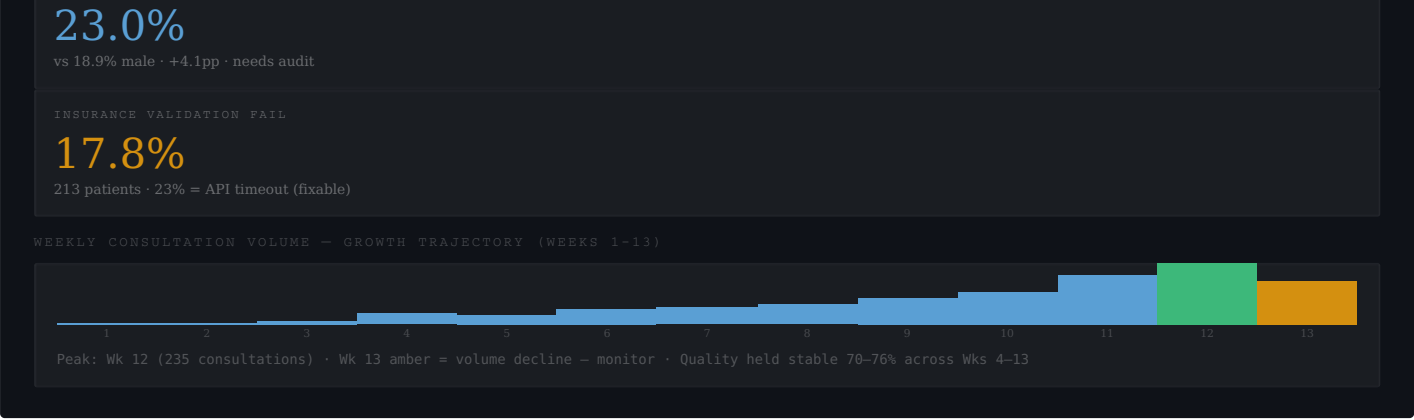
Target 1.0 · 60% enrolled vs 83% national

UNINSURED COMPLETION GAP

-21.6pp

Insured 73.1% · Uninsured 51.5%

FEMALE REFERRAL RATE



Dashboard Annotations

- A1 Safety Alert Strip (top row, red/amber)**
- Positioned first to enforce the committee's primary obligation — patient safety — before any operational metrics. A member who sees green KPIs before the lab review alert leaves with a false impression of platform safety. Decision: whether to escalate the unreviewed result issue to medical leadership immediately or defer to an agenda item.
- A2 Quality KPI Row (4 tiles — care episode chain)**
- Four tiles represent the end-to-end episode: Was consultation completed? Was diagnosis recorded? Was lab reviewed? Was medication dispensed? Together they form a quality chain; each answers a distinct governance question. The ICD tile's GP vs. Nurse split is retained because it directly identifies where to target the intervention. Decision: which stage of the care pathway has the weakest compliance.
- A3 Chronic Disease Follow-Up (three separate figures, not averaged)**
- Hypertension's 20.2% is clinically different from Diabetes's 29.9% — averaging them would obscure the worst-performing condition. The referral pipeline uses a three-segment bar because the governance question is proportional, not absolute. Decision: whether to set and enforce a follow-up rate target for chronic conditions in Month 4.
- A4 Equity Strip (four tiles — four distinct access barriers)**
- Equity belongs on a clinical governance dashboard, not only in funder reports — access inequality is a governance failure. Four metrics show four different barrier types: geographic (CPI), economic (uninsured gap), gender (referral), and infrastructure (API timeout). The CPI is shown as a decimal index to emphasise it has a target. Decision: whether to request an equity workplan and escalate the API timeout issue to MoH ICT.
- A5 Weekly Volume Sparkline (bottom)**
- Volume trend is included as a platform stability signal — governance needs to assess whether quality is holding as volume scales. The roughly stable completion rate (70–76%) across weeks 4–13 despite an 18× volume increase is a positive governance signal. Week 13 amber flags a dip requiring investigation. Decision: confirming quality has not degraded with growth.

METRIC EXCLUDED — LAB TURNAROUND TIME (TAT) BY FACILITY

Mean lab TAT (range: 35.2h–37.5h across four facilities, a 2.3h spread) was excluded because at this stage of platform maturity it is not discriminating — all facilities perform similarly. TAT is an **operational metric** (How fast is the lab?) not a safety or governance metric. It belongs in an *Operations & Lab Performance Report* for facility managers and supply chain teams. When TAT variation widens or a facility-specific pattern (high TAT + high non-review rate) emerges, it should be promoted to governance-level attention. Premature inclusion on the governance dashboard would dilute signal with noise.

TeleClinic Platform — Month 3 Learning Brief

Prepared for the Gates Foundation Programme Team. Period: February – April 2026. Classification: Quarterly Funder Brief.

Platform: TeleClinic (Rwanda) Period: Month 3 · Feb–Apr 2026 Data tables: 7 · Records: 4,766 Patients in scope: 1,200 registered

1. Platform Performance at Month 3

The platform is scaling and reaching patients across Rwanda's 12 districts, with strong and consistent growth. Consultation volume grew from 3 in Week 1 to 235 in Week 12 — a 78-fold increase within a single quarter — while the consultation completion rate held stable between 70% and 76% throughout, suggesting that quality has not degraded as volume grew. By Month 3, the platform had completed over 800 consultations, with respiratory infections (21%), malaria (16%), and the combined chronic disease burden of hypertension and diabetes (21%) representing the top conditions — a case mix consistent with Rwanda's real primary care burden, not an urban convenience caseload.

Clinical quality indicators show a platform that is functional but not yet consistently excellent. Clinicians entered ICD diagnostic codes in 79% of completed consultations and clinical notes in 85% — meaningful baselines for a three-month-old platform. However, one in five completed clinical encounters ends without a structured diagnosis code, compromising disease surveillance and insurance billing. Importantly, nurse clinicians outperform GP clinicians on documentation compliance (83.9% vs 77.2%), suggesting the gap is a workflow design issue — fixable without a system-wide intervention.

The platform is not yet delivering consistent continuity of care for its most clinically complex patients. Fewer than 30% of patients seen for diabetes, hypertension, or mental health received any structured follow-up contact within the quarter. For hypertension — Rwanda's leading cardiovascular risk factor — the follow-up rate was 20.2%. A telehealth platform that sees a hypertensive patient once but never monitors their blood pressure response to treatment is providing a transactional service, not care management. This is the programme's most important clinical quality challenge for the next quarter.

2. What We Cannot Yet Conclude

We cannot conclude the platform is improving patient health. The dataset captures every operational step but contains no measure of whether patients' conditions improved, stabilised, or worsened after consultation. TeleClinic can demonstrate it is seeing patients. It cannot yet demonstrate it is helping them. Before any impact claim is made to government partners or in published evaluations, a minimal outcome measure — even a structured clinician-assessed change-in-status at follow-up — is essential.

We cannot conclude the platform is equitable. The 22.9 percentage-point shortfall in rural enrolment relative to Rwanda's 83% rural population is a signal, not a conclusion. Without patient-level socioeconomic data (Rwanda's Ubudehe poverty classification) and data on booking attempts that failed before reaching a consultation, we cannot determine whether low rural enrolment reflects active exclusion, lower awareness, or supply-side capacity gaps.

We cannot conclude that the 17.8% registration failure rate reflects patient ineligibility. Approximately 23% of insurance validation failures were caused by API timeouts — a system error, not a patient eligibility issue. These patients may be eligible but have been functionally excluded. Until a provisional access pathway is implemented, failed validations cannot be reliably interpreted as "patient is uninsured."

3. One Recommended Change to the Monitoring System

The single most important change: **mandate clinician acknowledgement of lab results before a consultation episode can be closed.**

Currently, 62 uploaded lab results (17.7% of those successfully transmitted) were never opened by a clinician. The platform allows records to be marked complete — and new appointments booked — without the ordering clinician viewing a pending result. For a platform managing malaria, diabetes, and hypertension, this is a workflow design failure with direct patient safety consequences.

The fix is technically simple: when a lab result is uploaded to a completed consultation, trigger a mandatory in-platform notification and require an acknowledgement action before the system allows a new consultation for that patient. This generates an `acknowledged_datetime` field, creates an auditable review trail, and converts a silent failure mode into a visible, trackable clinical obligation. It costs minimal development effort, requires no patient-facing change, and if implemented before Month 4, would allow the programme to demonstrate measurable safety improvement within a single reporting period.

4. Priority Question for Months 4–6

THE PRIORITY QUESTION

Does receiving a TeleClinic consultation — compared to not receiving one — change the clinical trajectory of patients with chronic disease?

Every other question about platform performance is secondary to this one. Volume, completion rates, ICD compliance, and equity metrics all describe the platform's machinery. This question asks whether the machine produces anything clinically valuable. By Month 6, the

platform will have enough follow-up consultations to compare clinical indicators across at least two time points for its chronic disease cohort. The monitoring system should immediately begin capturing: (a) a structured outcome measure at each follow-up consultation — blood pressure reading, HbA1c if available, patient-reported symptom status; (b) a comparison denominator — patients with the same diagnosis who registered but did not complete a consultation. Even an imperfect internal comparison would allow the programme to move from "TeleClinic is operating" to "TeleClinic is working." Without this shift, the platform cannot defend its budget in a tighter fiscal environment, or iterate its clinical model based on evidence of what is — and is not — improving health.

If You Had to Choose One Metric as the Primary Indicator of Platform Success

North Star: Chronic Disease Follow-Up Completion Rate

PRECISE DEFINITION

The proportion of patients who completed a consultation for a chronic condition (diabetes, hypertension, or mental health) and subsequently completed at least one follow-up consultation within 90 days. Numerator: patients with ≥ 1 completed follow-up linked to a chronic disease consultation within the quarter. Denominator: all patients with a completed chronic disease consultation. Exclusions: patients who died, explicitly declined follow-up (if captured), or whose chronic diagnosis was incidental to an acute presentation. At Month 3: **26.2%** (Diabetes 29.9% · Hypertension 20.2% · Mental Health 28.6%).

WHAT IT CAPTURES THAT OTHER METRICS DON'T

Every other metric in this dataset measures a *process*: Did the consultation complete? Was the ICD code entered? Was the result reviewed? These are necessary but not sufficient measures of platform value. The chronic disease follow-up rate is the first metric that measures whether the platform is functioning as a *care system* rather than a *transaction system*.

Rwanda's chronic disease burden is structural. Hypertension, diabetes, and mental health conditions do not resolve after a single consultation — they require ongoing monitoring, treatment titration, and patient engagement over months and years. A telehealth platform that sees a diabetic patient once and then loses contact has provided a service, not care. The follow-up rate asks: does this patient come back? Does the clinician reach out? Is there a relationship forming, or just a series of disconnected encounters?

It also captures platform trust. A patient who returns for follow-up has found the experience worth repeating — it integrates satisfaction, accessibility, and perceived clinical value into a single behavioural signal. A patient who doesn't return may have recovered, but they may also have found the platform unreliable, inaccessible, or unhelpful. Distinguishing these requires additional data, but the rate itself anchors the investigation.

BLIND SPOTS AND LIMITATIONS

This metric has four meaningful limitations that must be tracked alongside it. First, it **misses acute care quality**: it says nothing about whether the 127 malaria patients or 169 respiratory infection patients were correctly treated, and a platform that excels in chronic disease management could still harm patients through poor acute clinical decisions.

Second, it is **vulnerable to gaming**: if clinicians schedule pro forma follow-up appointments that patients then fail to attend, the denominator grows without any genuine care engagement. The metric must track *completed* follow-ups, not merely scheduled ones.

Third, it cannot distinguish **clinical success from clinical failure**: a patient who follows up because their condition improved, and a patient who follows up because it worsened, both count as successes. Only when linked to a clinical outcome measure (blood pressure, HbA1c) does follow-up rate become interpretable. Without that linkage, it measures engagement, not health.

Fourth, it is **a lagging indicator**: a problem identified through the follow-up rate in Month 3 reflects decisions made in Month 1. Faster-moving safety signals — the lab result review rate, the ICD completion rate — are needed alongside it to detect emerging problems before they compound. The North Star orients the programme toward its ultimate purpose; the safety metrics keep it from drifting into harm while navigating there.

What I Prioritised, What I Left Out, Where I Felt Uncertain

Where I started: I read the data dictionary before writing any code, then loaded all seven tables and performed a full structural audit — shapes, dtypes, nulls, value counts, duplicate checks — before forming any analytical view. The 307 simultaneous null rows in Consultations were the first thing I flagged, and correctly identifying them as a denominator trap (not a data entry failure) shaped every subsequent metric calculation. Getting the denominator right was the single most important early decision.

What I prioritised: I prioritised analytical judgement over completeness. Rather than computing every possible metric, I asked which five metrics — if they could only report five — would give a clinical governance committee and a funder the clearest view of whether the platform is safe, equitable, and working. I also prioritised patient safety signals over operational metrics: the 62 unreviewed lab results are not the most complex finding, but they are the most urgent, and I made sure that urgency was communicated clearly in every relevant section rather than buried in a data table.

What I left out: I did not compute lab TAT by test type or by facility beyond summary statistics — the 2.3-hour spread across facilities is not yet discriminating and would have added bulk without insight. I did not compute prescription lag distributions beyond the headline dispensing rate. I also did not attempt age-stratified consultation rates, which would have been interesting but required assumptions about district-level age distributions I don't have. I flagged these as gaps rather than filling them with speculative analysis.

Where I felt uncertain: The join problem was the most challenging element to characterise precisely. All patient ids join cleanly — there is no broken key — so the "join problem" is logical, not referential. I was initially uncertain whether this was the intended finding or whether there was a second, more traditional orphan-record join problem elsewhere. After systematically checking every table pair (Consultations ↔ Patients, Follow-Ups ↔ Consultations, Labs ↔ Consultations, etc.) and finding clean joins throughout, I concluded the Musanze logical anomaly was the intended finding, and framed it accordingly. I also felt uncertain about the uninsured cohort: at 13.1% of registrants, it is plausibly under-counted (some patients may have been miscoded), and I noted this caveat explicitly rather than treating the figure as definitive.

On the North Star: I chose the chronic disease follow-up rate rather than a single safety metric because I believe the most important purpose of a telehealth platform in Rwanda's context is not process compliance — it is building longitudinal care relationships for conditions that cannot be resolved in a single encounter. I addressed its blind spots honestly because a North Star that can't be interrogated isn't useful. The lab result review rate is the most urgent metric right now. The chronic disease follow-up rate is the most important for assessing whether the platform succeeds in its stated purpose over time. Both are necessary; they answer different questions.

DATA SOURCE & CODE REPOSITORY

GitHub — TeleClinic Assessment

https://github.com/josegit94/teleclinic_Assessment

REPOSITORY CONTAINS

Analysis scripts · Dataset · Dashboard outputs · Full submission